

PostgreSQL - GROUPING SETS, ROLLUP & CUBE (Complete Notes)

Practice Dataset

```
CREATE TABLE orders (  
  id INT PRIMARY KEY,  
  continent VARCHAR(30),  
  country VARCHAR(30),  
  city VARCHAR(30),  
  amount INT );  
  
INSERT INTO orders VALUES  
(1, 'Asia', 'India', 'Bangalore', 1000),  
(2, 'Asia', 'India', 'Chennai', 2000),  
(3, 'Asia', 'Japan', 'Tokyo', 4000),  
(4, 'Asia', 'Japan', 'Hiroshima', 5000),  
(5, 'Europe', 'United Kingdom', 'London', 1000),  
(6, 'Europe', 'United Kingdom', 'Manchester', 2000),  
(7, 'Europe', 'France', 'Paris', 4000),  
(8, 'Europe', 'France', 'Cannes', 5000);
```

Cheat Sheet

1. Normal GROUP BY

```
By Continent  
SELECT continent, SUM(amount)  
FROM orders  
GROUP BY continent;  
  
By Country  
SELECT country, SUM(amount)  
FROM orders  
GROUP BY country;  
  
By City  
SELECT city, SUM(amount)  
FROM orders  
GROUP BY city;
```

2. GROUPING SETS

Think:
"I choose exactly what summaries I want."

Country + Grand Total

```
SELECT continent, country, SUM(amount)  
FROM orders  
GROUP BY GROUPING SETS  
(  
  (country),  
  ()  
);
```

Continent + Country

```
SELECT continent, country, SUM(amount)  
FROM orders  
GROUP BY GROUPING SETS
```

```

(
(continent),
(country)
);
Country + City + Grand Total

SELECT country,city,SUM(amount)
FROM orders
GROUP BY GROUPING SETS
(
(country),
(city),
()
);

Everything you choose

SELECT continent,country,city,SUM(amount)
FROM orders
GROUP BY GROUPING SETS
(
(continent),
(country),
(city),
()
);

```

Equivalent UNION ALL (How GROUPING SETS Works Internally)

The above GROUPING SETS query is equivalent to:

```

SELECT
    continent,
    NULL AS country,
    NULL AS city,
    SUM(amount)
FROM orders
GROUP BY continent

UNION ALL

SELECT
    NULL AS continent,
    country,
    NULL AS city,
    SUM(amount)
FROM orders
GROUP BY country

UNION ALL

SELECT
    NULL AS continent,
    NULL AS country,
    city,
    SUM(amount)
FROM orders
GROUP BY city

UNION ALL

SELECT
    NULL AS continent,
    NULL AS country,
    NULL AS city,
    SUM(amount)
FROM orders;

```

Explanations

```
(continent) -> Query grouped by continent
(country)   -> Query grouped by country
(city)      -> Query grouped by city
()          -> Grand Total UNION ALL combines the results into
```

a single output.

Remember:

- Every SELECT in a UNION ALL must return the same number of columns.
- NULL placeholders keep the column positions consistent.

3. ROLLUP

Think:

"Go from detailed to summarized."

```
SELECT continent, country, SUM(amount)
FROM orders
GROUP BY ROLLUP(continent, country);
```

Equivalent to

```
GROUPING SETS
(
  (continent, country),
  (continent),
  ()
)
```

Continent -> Country -> City -> Total

```
SELECT continent, country, city, SUM(amount)
FROM orders
GROUP BY ROLLUP(continent, country, city);
```

Equivalent to

```
(continent, country, city)
(continent, country)
(continent)
()
```

Notice:

ROLLUP removes one column at a time from RIGHT to LEFT.

4. CUBE

Think:

"Give me every possible subtotal."

```
SELECT continent, country, SUM(amount)
FROM orders
GROUP BY CUBE(continent, country);
```

Equivalent to

```
(continent, country)
(continent)
(country)
()
```

Three columns

```
SELECT continent, country, city, SUM(amount)
FROM orders
GROUP BY CUBE(continent, country, city);
```

Equivalent to

- (continent, country, city)
- (continent, country)
- (continent, city)
- (country, city)
- (continent)
- (country)
- (city)

() There are $2^3 = 8$ grouping sets.

Practice Questions

- Q1 Show total sales by continent.
- Q2 Show total sales by country.
- Q3 Show Continent totals, Country totals and Grand Total using GROUPING SETS.
- Q4 Show Continent, Country, City, Country Total, Continent Total and Grand Total using ROLLUP.
- Q5 Show every possible subtotal using CUBE.
- Q6 Rewrite ROLLUP using GROUPING SETS.
- Q7 Rewrite CUBE using GROUPING SETS.

One-Page Notes

GROUPING SETS
-----You choose
the summaries.

GROUPING SETS
(
(A),
(B),
(C),
()
) Equivalent to multiple GROUP BY queries combined with UNION

ALL. -----

ROLLUP
-----Hierarchy.

ROLLUP (A, B, C)

Equivalent to

- (A, B, C)
- (A, B)
- (A)

() Removes columns from RIGHT to LEFT. -----

CUBE
---Every possible
combination.

CUBE (A, B, C)
Equivalent to

- (A, B, C)
- (A, B)
- (A, C)
- (B, C)

- (A)
- (B)
- (C)
- ()

Number of grouping sets = 2^n -----

-

Relationship

ROLLUP $\subset$ GROUPING SETS

CUBE $\subset$ GROUPING SETS

GROUPING SETS is the most flexible.

ROLLUP and CUBE are shortcuts.